

# 1. Introduction

This research investigates the application of modern machine learning techniques in industrial fault diagnosis. With the advancement of Industry 4.0, equipment condition monitoring and fault prediction have become increasingly critical. This research investigates the application of modern machine learning techniques in industrial fault diagnosis. With the advancement of Industry 4.0, equipment condition monitoring and fault prediction have become increasingly critical. This research investigates the application of modern machine learning techniques in industrial fault diagnosis. With the advancement of Industry 4.0, equipment condition monitoring and fault prediction have become increasingly critical.

## 1.1 Background

This research investigates the application of modern machine learning techniques in industrial fault diagnosis. With the advancement of Industry 4.0, equipment condition monitoring and fault prediction have become increasingly critical.

### 1.1.1 Historical Context

This research investigates the application of modern machine learning techniques in industrial fault diagnosis. With the advancement of Industry 4.0, equipment condition monitoring and fault prediction have become increasingly critical.

## Chapter 2 Methods

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## 2.1 Experimental Design

This research investigates the application of modern machine learning techniques in industrial fault diagnosis. With the advancement of Industry 4.0, equipment condition monitoring and fault prediction have become increasingly critical.

### 2.1.1 Data Collection

This research investigates the application of modern machine learning techniques in industrial fault diagnosis. With the advancement of Industry 4.0, equipment condition monitoring and fault prediction have become increasingly critical.